University of Florida, Department of Mathematics

Analysis PhD Exam, October 1994

(X, Σ, μ) is a finite measure space.

I.

State the following theorems:

1. Lebesgue dominated convergence theorem
2. Fatou Lemma
3. Vitali theorem
4. Monotone convergence theorem
5. The Radon–Nikodym theorem
6. The Egorov theorem
7. The Fubini theorem
8. The Lebesgue decomposition theorem
9. The Uniform boundedness theorem
10. The Hahn–Banach theorem

II.

1. Prove the $(L^1)^* = L^\infty$

III. Problems

1. Find $\lim_{n\to\infty} \int_0^1 nx^n \, dx$. (Hint: use the monotone convergence theorem)

2. Let μ be the Lebesgue measure on $\mathbb{R}$ and f a Lebesgue-integrable function on $\mathbb{R}$, such that $\int_0^x f \, d\mu = 0$ for every $x \in \mathbb{R}$. What can you say about f?

3. **Warning: Suspected error.** The source statement was retained, but on a general finite measure space the displayed hypotheses do not imply the stated lower bound. An additional normalization such as $\mu(X) = 1$, or a different lower bound, may have been intended; the correction is not uniquely determined.

   Let $f \geq 0$ be μ-integrable such that $\int f \, d\mu \leq 1$. Prove that $\int \frac{1}{f} \, d\mu \geq 1$. (Hint, use the Schwarz inequality for $\sqrt{f}$ and $\frac{1}{\sqrt{f}}$.)

4. Let R be a ring of subsets of X, E a Banach space and $\mu : R \to E$ be a finitely additive measure. Prove that μ is σ-additive iff for any decreasing sequence (A_n) from R with $\bigcap_n A_n = \phi$ we have $\lim \mu(A_n) = 0$.